THE HAPPINESS ISSUE

CONDUCTING A TECH SYMPHONY

Roll over silicon. Gallium Nitride is fast emerging as the **go-to semiconductor compound in power-electronics** and wireless communication. In India it's being made at AGNIT

By AJAY SUKUMARAN

Have you ever wondered how the size of your smartphone has grown bigger over the years but the fast charger that keeps it going is still a compact little device? Thank Gallium Nitride (GaN). It's a semiconductor compound more efficient than silicon when it comes to powering electronics—that means anything to do with controlling and converting the flow of electricity into an electronic device, be it a phone, laptop or an electric vehicle. GaN operates at a higher frequency than silicon. There's also less wastage of energy: the old chargers used to run 5-10 watts; fast chargers nowadays use 30-60 watts in practically the same footprint, or even smaller. Now, a start-up, AGNIT Semiconductors, India's first private sector company in semiconductor manufacturing, is aiming for a slice of the market for GaN components.

Where else does GaN make a difference? In white light-emitting LEDs and radio frequency (RF) electronics. The latter covers the entire span of wireless communication, from radars to 5G phone connectivity. "Gallium Nitride is going to play a very important role in the way we communicate wirelessly and consume energy in the form of electricity," says Prof. Srinivasan Raghavan, chairman of the Centre for Nano Science and Engineering (CeNSE) at the Bengaluru-based Indian Institute of Science (IISc). The centre hosts India's largest academic fabrication facility, or fab. Over the past 18 years, about 10 faculty members from across three depart-



MASTER CONDUCTORS (From left) R. Muralidharan, Srinivasan Raghavan and Hareesh Chandrasekar